USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.PROT.SRC3.14 Source Info Update
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Revise step 4.k.3 check which compares the PDP of the Source Capabilities from the current Explicit contract with Port Guaranteed PDP to avoid false failures of compliant EPR Sources operating in SPR mode. Note: Changes added to the approved version of the ECR https://groups.usb.org/wg/pdcompli/document/18924. Changes for the proposed version highlighted in blue.

Benefits as a result of the proposed changes:
Better alignment to the base spec definition – no power rule on current contract(or =Reported PDP) = Port Guaranteed PDP, eliminating false failures for EPR-capable Sources.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Update test implementation per ECR update

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

TEST.PD.PROT.SRC3.14 Source Info

To Text:

TEST.PD.PROT.SRC3.14 Source Info

Description: As a Sink, the Tester verifies that the UUT replies to the *Get_Source_Info* Message with Source_Info.

Test Specific Tester Behavior: N/A

Test Conditions:

	Consumer Only	Provider Only, C/P, P/C, DRP
Rev3ChkdSrc		✓
Rev3UnchkdSrc		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester sends the *Get_Source_Info* Message to the UUT.
3. The Tester checks the response of the *Get_Source_info* [TEST.PD.PROT.SRC3.14#2]:
    a. If the UUT responds with a *Not_Supported* Message, the Tester checks Has_Invariant_PDOs VIF field is set to Yes and EPR_Supported_As_Src VIF field is set to No and check number of SRC+DRP VIF objects is equal to 1 (Use Type_C_State_Machine VIF field to find the number of SRC/DRP VIF objects) and the test stops here.
    b. Else, the check fails if the UUT does not reply with the *Source_Info* Message, and the test stops here.
4. The Tester checks the response, and the check fails if one of the following conditions is met: [TEST.PD.PROT.SRC3.14#1]
    a. The Number of Data Objects in the *Source_Info* Message is not 2.
    b. In SIDO1 and SIDO2, B31 (Port Type) does not match Port_Managed_Guaranteed_Type VIF field.
    c. In SIDO2, if B30 (DPS Port) is 1, Bit31 (Port Type) is not 0.
    d. In SIDO2, B30 (DPS Port) does not match to the DPS_Supported VIF field.
    e. In SIDO1, B30…24 are not Zeros.
    f. In SIDO2, B29…18 are not Zeros.
    g. In SIDO1, B23…16 (Port Maximum PDP) is not equal to the round down value of PD_Power_As_Source VIF field/1000.
    h. In SIDO1, B15…8 (Port Present PDP) is higher than Maximum PDP
    i. In SIDO1, B7…0 (Port Reported PDP) is higher than Maximum PDP or not the computed PDP of *Source Capabilities* from the current Explicit Contract. The PDP of the *Source Capabilities* computed as the integer part of the product of the highest-voltage fixed PDO voltage times its maximum current.
        - Reported PDP = Integer of VxI, where V = (Voltage, B19…10)*50mV and I = (Maximum Current, B9…0)*10mA, taken from the highest-voltage Fixed Supply PDO returned in the *Source Capabilities* Message

j. In SIDO2, B8…0 (Port Guaranteed PDP) does not match the PD_Power_As_Source_Guaranteed VIF field

k. If the Port_Managed_Guaranteed_Type VIF field is set to 1 (Guaranteed),

1. In SIDO1, B23…16 (Port Maximum PDP) is not equal to B15…B8 (Port Present PDP)
2. In SIDO2, B17…9 (Port Maximum PDP) is not equal to B8…0(Port Guaranteed PDP)
3. If EPR_Supported_As_Src VIF field is set to No, the PDP of the *Source Capabilities* from the current *Explicit Contract* is less than SIDO2 B8…0 (Port Guaranteed PDP)
4. If EPR_Supported_As_Src VIF field is set to Yes, the PDP of the *Source Capabilities* from the current *Explicit Contract* is
   a. greater than SIDO2 B8…0(Port Guaranteed PDP) and
   b. less than the product of the highest-voltage Fixed Supply PDO ≤ 20V in the VIF SrcPdoList: Src_PDO_Voltage *Src_PDO_Max_Current

5. The Tester disconnects from the UUT and waits for $tV_{BUS}OFF$ (650ms,Type-C spec).
6. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7
   b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1
   c. During bring-up, if SOP' *Discover Identity* Command is received, the Tester replies as Passive Cable with V_{BUS} Current Handling Capability set to 10b(5A), Maximum V_{BUS} Voltage set to 00b(20V) and EPR Mode Capable Field set to 0b (EPR Mode Capable)
7. The Tester sends the *Get_Source_Info* to the UUT.
8. The Tester checks the response of *Get_Source_info* [TEST.PD.PROT.SRC3.14#3] :
   a. If the UUT responds with a *Not_Supported* Message, the Tester checks Has_Invariant_PDOs VIF field is set to Yes and EPR_Supported_As_Src VIF field is set to No, and check number of SRC+DRP VIF objects is equal to 1(Use Type_C_State_Machine VIF field to find the number of SRC/DRP VIF objects) and stops here.
   b. Else, the check fails if the UUT does not reply with the *Source_Info* Message and the test stops here.
9. The Tester checks the response, and the check fails if one of the following conditions is met in the *Source_Info* Message [TEST.PD.PROT.SRC3.14#4]:
   a. The Number of Data Objects in the *Source_Info* Message is not 2.
   b. In SIDO1 and SIDO2, B31 (Port Type) does not match Port_Managed_Guaranteed_Type VIF field.
   c. In SIDO2, if B30 (DPS Port) in SIDO2 is 1, Bit31 (Port Type) is not 0.
   d. In SIDO2, B30 (DPS Port_ does not match to the DPS_Supported VIF field
   e. In SIDO1, B30…24 are not Zeros.
   f. In SIDO2, B29…18 are not Zeros.
   g. In SIDO1, B23…16 (Port Maximum PDP) is not equal to the round down value of PD_Power_As_Source VIF field/1000
   h. In SIDO1, B15…8 (Port Present PDP) is higher than Maximum PDP
   i. In SIDO1, B7...0 (Port Reported PDP) is higher than Maximum PDP or not the computed PDP of *Source Capabilities* from the current Explicit Contract. The PDP of the *Source Capabilities* computed as the integer part of the product of the highest-voltage fixed PDO voltage times its maximum current..
      - Reported PDP = Integer of VxI, where V = (Voltage, B19…10)*50mV and I = (Maximum Current, B9…0)*10mA, taken from the highest-voltage Fixed Supply PDO returned in the *Source Capabilities* Message

j. In SIDO2, B8…0 (Port Guaranteed PDP) does not match PD_Power_As_Source_Guaranteed VIF field.
k. If VIF field Captive_Cable = No, the Port_Managed_Guaranteed_Type VIF field is set to 1 (Guaranteed) and EPR_Supported_As_SRC is Yes, B23..16 (Port Maximum PDP) is equal to B15..8 (Port Present PDP) in SIDO1.
l. If VIF field Captive_Cable = YES and the Port_Managed_Guaranteed_Type VIF field is set to 1 (Guaranteed) and EPR_Supported_As_SRC is Yes, and B23..16 (Port Maximum PDP) is not equal to B15..8 (Port Present PDP) in SIDO1
m. If VIF field Captive_Cable = No, the PD_Power_as_Source VIF field is greater than 100W and the Port_Present in SIDO1 is higher than 100W.

10. If VIF field is Captive_Cable = Yes, the test stops here.
11. The Tester disconnects from the UUT and waits for *$tV_{BUS}OFF$*.
12. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1
    c. During bring-up, if the SOP' *Discover Identity* Command is received, the Tester intentionally does not reply with the *GoodCRC* Message.
13. The Tester sends the *Get_Source_Info* Message to the UUT.
14. The Tester checks the response of the *Get_Source_Info* Message [TEST.PD.PROT.SRC3.14#5].
    a. If the UUT responds with a Not_Supported Message, the Tester checks Has_Invariannt_PDOs VIF field is set to Yes and EPR_Supported_As_Src VIF field is set to No and check number of SRC+DRP VIF objects is equal to 1 (Use Type_C_State_Machine VIF field to find the number of SRC/DRP VIF objects) and test stops here.
    b. Else, the check fails if the UUT does not reply with the *Source_Info* Message and the test stops here.
15. The Tester checks the response, and the check fails if one of the following conditions is met in the *Source_Info* Message [TEST.PD.PROT.SRC3.14#6].
    a. The number of Data Objects in the Source_Info Message is not 2.
    b. In SIDO1 and SIDO2, B31 (Port Type) does not match Port_Managed_Guaranteed_Type VIF field. If B30 (DPS Port) in SIDO2 is 1, Bit31 (Port Type) is not 0.
    c. In SIDO2, B30 (DPS Port) does not match the DPS_Supported VIF field.
    d. In SIDO1, B30…24 are not Zeros.
    e. In SIDO2, B29…18 are not Zeros.
    f. In SIDO1, B23…16 (Port Maximum PDP) is not equal to the round down value of PD_Power_As_Source VIF field/1000.
    g. In SIDO1, B15…8 (Port Present PDP) is higher than Maximum PDP.
    h. In SIDO1, B7…0 (Port Reported PDP) is higher than Maximum PDP or not the computed PDP of *Source Capabilities* from the current Explicit Contract. The PDP of the *Source Capabilities* computed as the integer part of the product of the highest-voltage fixed PDO voltage times its maximum current..
        - Reported PDP = Integer of VxI, where V = (Voltage, B19…10)*50mV and I = (Maximum Current, B9…0)*10mA, taken from the highest-voltage Fixed Supply PDO returned in the *Source Capabilities* Message
    i. In SIDO2, B8…0 (Port Guaranteed PDP) does not match to the PD_Power_As_Source_Guaranteed VIF field.
    j. If the Port_Managed_Guaranteed_Type VIF field is set to 1 (Guaranteed) and EPR_Supported_As_SRC is Yes, and B23…16 (Port Maximum PDP) is equal to B15…8 (Port Present PDP) in SIDO1.
    k. If B23…16 (Port_Maximum_PDP) is greater than 60W and B15...8 (Port_Present_PDP) is equal to Port_Maximum_PDP in SIDO1.